EXPERIMENT 22
Reference Resolution in Text

Aim

To implement a simple reference-resolution program that links pronouns to likely preceding nouns.

Procedure

1. Tokenize a short text into sentences and words.
2. Collect candidate nouns from the text.
3. For each pronoun, find the nearest preceding candidate noun.
4. Display the pronoun and its proposed referent.
5. Note that this is a simplified heuristic.

Program

```
import re

text = "Ravi bought a laptop. He uses it every day."
sentences = re.split(r"(?<=[.!?])\s+", text)
candidates = []

for sentence in sentences:
    words = re.findall(r"\b[A-Za-z]+\b", sentence)
    for word in words:
        if word.lower() not in {"he", "she", "it", "they", "a", "the", "uses", "bought",
"every", "day"}:
            candidates.append(word)
    for pronoun in [w for w in words if w.lower() in {"he", "she", "it", "they"}]:
        referent = candidates[-1] if candidates else "unknown"
        print(pronoun, "->", referent)
```

Output

```
OUTPUT — EXPERIMENT 22
He -> laptop
it -> laptop
(Note: this simple heuristic can be wrong; it is not a full coreference
    system.)
```

Result

Thus, a basic heuristic was used to propose referents for pronouns in the text.